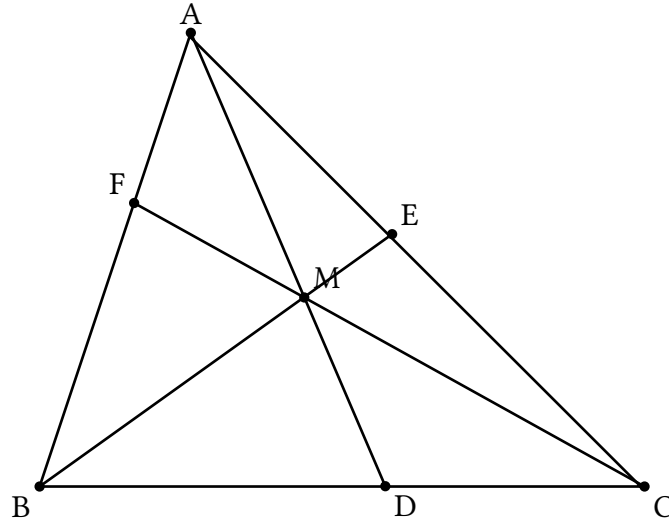


Easy Math Document

Point M inside triangle ABC



Let M be a point inside triangle ABC. The rays AM, BM, CM meet BC, CA, AB at D, E, F respectively.

a) Prove $\frac{MD}{AD} + \frac{ME}{BE} + \frac{MF}{CF} = 1$

Triangles MBC and ABC share the base BC. Since A, M, D are collinear, the distances from M and A to BC are proportional to MD and AD:

$$\frac{MD}{AD} = \frac{[MBC]}{[ABC]}$$

Similarly:

$$\frac{ME}{BE} = \frac{[MCA]}{[ABC]}, \quad \frac{MF}{CF} = \frac{[MAB]}{[ABC]}$$

Adding them, since triangles MBC, MCA, MAB exactly compose triangle ABC:

$$\frac{MD}{AD} + \frac{ME}{BE} + \frac{MF}{CF} = \frac{[MBC] + [MCA] + [MAB]}{[ABC]} = 1$$

b) Prove $\frac{AM}{AD} + \frac{BM}{BE} + \frac{CM}{CF} = 2$

Since M lies between A and D, $AM + MD = AD$, so:

$$\frac{AM}{AD} = 1 - \frac{MD}{AD}$$

Similarly for B and C. Adding them and using part a):

$$\frac{AM}{AD} + \frac{BM}{BE} + \frac{CM}{CF} = 3 - 1 = 2$$

c) At least one ratio ≥ 2 and at least one ratio ≤ 2

Set $x = \frac{MD}{AD}$, $y = \frac{ME}{BE}$, $z = \frac{MF}{CF}$. By part a): $x + y + z = 1$.

We have:

$$\frac{AM}{MD} = \frac{1-x}{x}, \frac{BM}{ME} = \frac{1-y}{y}, \frac{CM}{MF} = \frac{1-z}{z}$$

- Suppose all three ratios are < 2 . Then $\frac{1-x}{x} < 2 \Rightarrow x > \frac{1}{3}$, similarly $y > \frac{1}{3}$ and $z > \frac{1}{3}$. Hence $x + y + z > 1$, a contradiction.
- Suppose all three ratios are > 2 . Then $x, y, z < \frac{1}{3}$, so $x + y + z < 1$, also a contradiction.

Therefore among the three ratios there is always at least one ratio ≥ 2 and at least one ratio ≤ 2 .